

## COMMON COURSE ASSIGNMENT FORMAT

### A. Assignment Information

| Particular                    | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Department:                   | Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Programme:                    | B.E / B.Tech                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Course Code & Course Name     | UBA06__ / Applied Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Academic Year / Batch         | 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Faculty Name                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Assignment Title              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Date of Issue                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Date of Submission            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Maximum Marks                 | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Course Outcome(s) – CO        | CO1-CO5/ On successful completion of this assignment set, the student will be able to <b>APPLY</b> integral theorems in vector calculus to evaluate line and surface integrals over open and closed surfaces; <b>APPLY</b> Fourier series to represent periodic functions by distinguishing even and odd functions; <b>COMPUTE</b> Laplace transforms and <b>EMPLOY</b> them to solve initial value problems arising in engineering applications; <b>EMPLOY</b> Fourier transform techniques and Parseval's identity to analyze and interpret engineering signals and functions; and <b>ANALYZE</b> continuous-time and discrete-time systems using Z-transforms for system representation and response analysis — all within a single realistic engineering scenario per problem. |
| Bloom's Taxonomy Level        | L4 – Analyze / L5 – Evaluate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| SDG Mapping (SDG 1-17)        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Industry / Societal Relevance |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### B. Assignment Problem

The assignment should preferably require students to:

## **Problem Statement**

**Problem / Case / Design Challenge: Integrated Applied Vector Calculus and Transform Techniques**

## **C. Requirements and Constraints**

### ***Requirements***

- Formulate the selected engineering problem mathematically.
- Verify the analytical results using an independent mathematical, numerical, or computational method.
- Develop a program in Python or MATLAB to verify the manual/analytical calculations.
- Compare results, interpret the engineering meaning, make a justified conclusion.

### ***Constraints***

- Must formulate appropriate assumptions, initial/boundary conditions and constraints.
- Select and justify a suitable mathematical technique.
- Provide a manual/analytical solution; and independently verify the result using Python or MATLAB.
- Computational tools may be used for numerical calculations, visualization, and simulation but must not replace mathematical reasoning.
- Numerical accuracy, approximation and convergence issues must be addressed, and the final results must be interpreted in the engineering context to support an evidence-based conclusion.

## **D. Student Work**

### **1. Problem Understanding and Formulation**

Explain:

- What is the problem?
- What are the expected outcomes?
- What information/data is available?
- What assumptions are made?
- What constraints must be satisfied?

### **2. Application of Course Knowledge**

Identify and apply the relevant:

- Principles
- Theories
- Mathematical models
- Engineering concepts

Show necessary calculations, derivations or reasoning.

### **3. Solution / Design / Methodology**

Develop the proposed solution. Depending upon the nature of the course, this may include:

- Model
- Process
- Data analysis
- Mathematical solution

Where appropriate, consider **more than one possible solution**.

### **4. Use of Modern Tools**

Use appropriate tools wherever relevant.

Examples:

**Mathematics: Python / Matlab**

Students shall provide appropriate evidence of tool usage and outputs.

### **5. Results and Validation**

Present the results using appropriate:

- Tables
- Simulation outputs
- Experimental observations
- Screenshots
- Performance metrics
- Statistical analysis

Validate whether the proposed solution satisfies the stated requirements.

### **6. Analysis and Engineering Decision**

Interpret the results.

Where applicable:

- Compare alternatives.
- Identify advantages and limitations.

- Explain why the final solution was selected.
- Justify the decision using quantitative or qualitative evidence.

## 7. Broader Considerations

Where relevant to the problem, discuss the impact of the proposed solution with respect to:

- Sustainability

## 8. Conclusion

Summarize:

- Proposed solution
- Major findings
- Achievement of requirements
- Limitations
- Possible improvements

## 9. Student Reflection

- What did you learn from this assignment beyond what was directly taught in the classroom?
- If you were given additional time/resources, what would you improve and why?

## 10. References

Use an appropriate citation format and acknowledge all external sources, datasets, standards, software and AI-assisted tools used.

## E. Common Assessment Rubric

| Assessment Criterion                                                   | Marks      |
|------------------------------------------------------------------------|------------|
| Problem understanding & formulation                                    | 10         |
| Application of course/domain knowledge                                 | 20         |
| Solution methodology / design / implementation                         | 20         |
| Use of appropriate modern tools / techniques                           | 10         |
| Results, testing & validation                                          | 15         |
| Analysis, trade-offs & justification                                   | 15         |
| Broader considerations / professional responsibility, where applicable | 5          |
| Technical documentation & reflection                                   | 5          |
| <b>TOTAL</b>                                                           | <b>100</b> |

## F. CO–PO–Assessment Mapping

| Assessment Component               | CO       | PO(s)    | Bloom's Level | Marks |
|------------------------------------|----------|----------|---------------|-------|
| Problem formulation                | CO1      | PO1, PO2 | L3/L4         | 10    |
| Application of knowledge           | CO1, CO2 | PO1, PO2 | L3/L4         | 20    |
| Solution / Design / Implementation | CO1, CO2 | PO1, PO2 | L4/L5/L6      | 20    |
| Modern tool usage                  | CO2      | PO1      | L3/L4         | 10    |
| Validation                         | CO1, CO2 | PO2      | L4/L5         | 15    |
| Analysis & justification           | CO1, CO2 | PO2      | L4/L5         | 15    |
| Broader considerations             | CO2      | PO2      | L4/L5         | 5     |
| Documentation & reflection         | CO1, CO2 | PO2      | L3/L4         | 5     |

**Note:** Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO**.

## G. Faculty Evaluation & Continuous Improvement

### CO Attainment

| Performance Level | Criterion   |
|-------------------|-------------|
| Level 3           | $\geq 70\%$ |
| Level 2           | 60–69%      |
| Level 1           | 50–59%      |
| Level 0           | $< 50\%$    |

**Target CO Attainment:** \_\_\_\_\_

**Actual CO Attainment:** \_\_\_\_\_

### Faculty Analysis

**Areas in which students performed well:**

**Areas requiring improvement:**

**Corrective / Improvement Action:**

**Action to be implemented in the next offering:**

**Documents to be maintained**

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action